

## Zoho Level-2 Questions (Future Pattern)

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### 1. Numbers – Modified Digit Rearrangement

**Problem:**

Given a number, rearrange its digits to form the **next greater number** using the same digits. If no such number exists, print **-1**.

(Similar to "next permutation" but numeric specific.)

**Input:**

1234

**Output:**

1243

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### 2. Numbers – Balanced Number Check

**Problem:**

A number is **balanced** if the sum of digits on the left half = sum of digits on the right half.

Write a program to check whether a number is balanced.

**Input:**

123321

**Output:**

Balanced

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### 3. Strings – Smallest Window with All Characters

#### Problem:

Given a string **s**, find the **smallest substring** that contains **all distinct characters** of **s**.

#### Input:

aabcbcd bca

#### Output:

dbca

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### 4. Strings – Minimum Character Removal to Make Palindrome

#### Problem:

Find the **minimum number of characters** that need to be removed from a string to make it a palindrome.

#### Input:

abcda

#### Output:

2

#### Constraints:

- $1 \leq |s| \leq 1000$
  - Optimal DP solution expected.
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## 5. Arrays – Maximum Sum Subarray with Twist

### Problem:

Find the **maximum subarray sum** but with the condition that you **must exclude exactly one element**.

### Input:

[1, -2, 0, 3]

### Output:

4 // (1 + 0 + 3)

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## 6. Arrays – Rearrange by Frequency

### Problem:

Given an array, rearrange elements in descending order of frequency. If two numbers have the same frequency, the smaller number comes first.

### Input:

[2, 3, 2, 4, 5, 12, 2, 3, 3, 3, 12]

### Output:

3 3 3 3 2 2 2 12 12 4 5

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## 7. String + Array – Substring Anagram Indices

### Problem:

Given a string `s` and a pattern `p`, return all starting indices in `s` where the substring is an **anagram** of `p`.

### Input:

```
s = "cbaebabacd"  
p = "abc"
```

### Output:

```
[0, 6]
```

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## 8. String + Array – Merge Two Strings Alternately

### Problem:

Given two strings, merge them alternately. If one string is longer, append the rest.

### Input:

```
s1 = "abc"  
s2 = "pqrstu"
```

### Output:

```
apbqcrstu
```